

Chapter 4

Methods for Verifying Medical Device Cleanliness

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1 BACKGROUND

Cleaning can ordinarily be defined as the removal of unwanted contaminants deposited on or attached to the surface of a product. Cleaning may be done to allow adequate performance or to insure critical safety of the product. Confirming that products and their component parts are adequately cleaned is especially crucial for the medical device industry. The result of inadequate or improperly cleaned medical device products can be devastating causing harm to patients resulting in considerable morbidity and even mortality. While health and safety of patients are the primary concern, having unclean medical devices can have catastrophic business consequences as well, resulting in litigations and even ultimately business failure or closure.

To most patients, the cleanliness of medical devices begins and ends at whether or not the “things” about to enter their bodies are free from “germs.” But to the medical device manufacturer, cleanliness is a far more complex issue. Not only is there concern about microbial contamination, there is also the need to eliminate contaminants originating from manufacturing and sterilization processes.

Many, if not most, medical devices are sterile products. Consequently, a product labeled as sterile is something that the United States Food and Drug Administration (FDA) or any European regulatory agency is going to see as possessing a high level of risk to the public. As a result, the control of potential sources of contamination has been and continues to be an issue of primary importance. During the past several years, there has been escalating attention aimed at the development of better cleaning processes and procedures for medical devices. With the introduction of next-generation microporous metal and ceramic components (e.g., for hip and knee implants) current vapor degreasing or solvent fill-and-flush procedures that were once acceptable, may not satisfactorily penetrate pores and extract machining and grinding oils used in component manufacturing. Polymeric devices such as hydrocephalus shunts, aortic grafts, urinary and endotracheal catheters, etc., represent an especially demanding class of biomaterials. They cannot be cleaned or extracted with traditional methods because of the absorption of solvent by the device material, making the solvent itself another source of contamination. The organic liquid solvent residues may be removed by applying heat, but the heating process can be destructive to the properties and performance of the part, potentially making the device unsafe. In this latter situation, an issue of material compatibility with the cleaning solvent becomes another key concern as well.

In today's highly regulated environment, medical device manufacturers must, and for the most part do, realize the significant role that cleaning methods can play in the consistency and quality of medical devices. While the health and safety of their patients may be foremost on their minds, performance, once it is in a health care provider's hands, also becomes a prominent issue. Although it is sometimes an afterthought, cleaning should be considered in every step of medical device design and manufacture. Manufacturers must ultimately use and depend upon cleaning systems to obtain the best and most consistent performance out of a device. These systems also ensure that the device is sterile and ready for use once it reaches a physician, nurse practitioner, or dentist.

Medical devices must be properly cleaned to ensure the removal of product residues, cleaning chemical residues, and microbes prior to manufacturing and packaging. Because cleaning methods are developed and validated to prevent the risk of producing contaminated products by confirming that the cleaning process is sufficient, it is important to establish method limits and select the proper cleaning techniques and detection methods. Therefore, it is important to test parts once the cleaning process and sterilization is complete to ensure

that they have been cleaned adequately. Validation and revalidation are key processes when establishing cleaning methods.

Once the cleaning, disinfecting process, or sterilization process has been implemented and validated, it becomes necessary to demonstrate that chemicals used in these processes are not left on the medical device or equipment at a level that may cause harm to medical personnel and to patients either as a single one-time exposure or multiple exposures from repeated use of a medical device. One of the first steps in the development of an effective cleaning validation method is the determination of necessary limits [1]. The FDA Guide to Inspections Validation of Cleaning Processes [2], for example, states, “The firm’s rationale for the residue limits established should be logical, based on the manufacturer’s knowledge of the materials involved and be practical, achievable, and verifiable.”

This chapter will examine various kinds of contaminants associated with medical devices and then explore various analytical techniques to detect their presence or absence. Another feature included in this chapter is the use of toxicological risk assessment as a way to set residue limits and as a method to evaluate the overall biological safety of any remaining or detected contaminants left on medical devices.

2 CLEANING ISSUES AND CONCERNS WITH MEDICAL DEVICES

To most patients, the cleanliness of medical devices begins and ends at whether or not the medical device or surgical instrument their doctor is about to use is free from “germs.” But to the medical device manufacturer, when the subject of medical device cleaning is considered, the main focus is on sterility and biocompatibility, including particulate analysis, bioburden, and cytotoxicity testing. While contaminants can be both viable (living) and nonviable (from nonliving sources), sterility, bioburden, and cytotoxicity testing will not address all the concerns regarding the questions of cleanliness of medical device products.

Technically, bioburden is a measure of an object’s contamination with microorganisms. Bioburden measures the living bacteria inhabiting a surface that has not been sterilized. The presence of microorganisms on a medical device can also lead to the presence of endotoxins. Both can present challenges in delivering a sterile and pyrogen-free medical device to the end-user [3].

Many conditions can cause bioburden but fundamentally, water is a growth medium for bacteria. Therefore, removing water from the cleaning process removes a growth medium for bioburden. This is why solvents are often preferable to aqueous (water-based) cleaners, because they present an environment that is hostile to bacteria growth. If bioburden is not addressed, it can lead to infection when that device is used in a patient. To address this potential concern, a manufacturer can specify a solvent-based process with submicrometer filtration [3].

What is forgotten many times is the effect of the manufacturing process and the potential for subsequent residual contaminants. These contaminants may include machining oils, lubricants, mold-release agents, cleaning solvents, and other process aids; all of which may have a negative impact on the performance of the medical device or put the health and safety of the patient at risk.

A single device may be exposed to many contaminants during manufacturing and cleaning. How a manufacturer deals with this issue can be quite variable. That is, how does one demonstrate the identity of the contaminants (soils), what are the levels of contaminants, what solvents or processes remove the contaminants, and finally how can one demonstrate that the whole process is reliable and repeatable? This is where the process of validation becomes part of the overall process [4]. How a manufacturer validates a cleaning operation is affected by the number and type of contaminants. In the medical device manufacturing industry, cleaning validation is generally performed by examining the finished device rather than the process equipment used to manufacture the device. Residues may be analyzed by direct surface analysis or measured after extraction. Extraction analysis is most common so the methods discussed in this chapter are extraction-based. In cleaning validation, residue acceptance criteria must be set. Acceptance criteria are generally set based on potential for the residue to affect biocompatibility, toxicity, or functionality of the finished device [4].

Contaminants usually fall into one of three categories: water-soluble and non-water-soluble residues, and nonsoluble debris. Water-soluble residues are usually ionic compounds such as detergents and salts. Non-water-soluble residues, such as oils, greases, and other hydrocarbons, are soluble in solvents other than water. Nonsoluble debris may include metals, organic and inorganic solids, and ceramics [5].

While there are certainly justifiable concerns about microbial contamination, there are also real concerns for manufacturing residues and the need to eliminate those contaminants originating from manufacturing processes. These contaminants include, but are not necessarily limited to, cutting or polishing fluids used in machining, adhesives used to join polymer subassemblies, mold-release agents, polymer-processing aids, carbon residues from laser drilling and cutting, and airborne contamination from the factory environment [4]. While this list is not all inclusive, one can clearly see that there are several types of contaminants that can be present on medical device surfaces that may be undesirable for product performance or safety (Fig. 4.1).

Cleaning is part of almost all manufacturing operations. Parts may require cleaning before they can be electroplated or painted, before they can be soldered or solvent welded, or before they can be packaged and shipped for end use. Thus, cleaning is necessary for various reasons to assure desirable product appearance, performance and safety. The presence of contaminants on a finished medical device can negatively impact its biocompatibility, resulting in adverse biological reactions such as inflammation, infection, and systemic allergic responses in some patients.

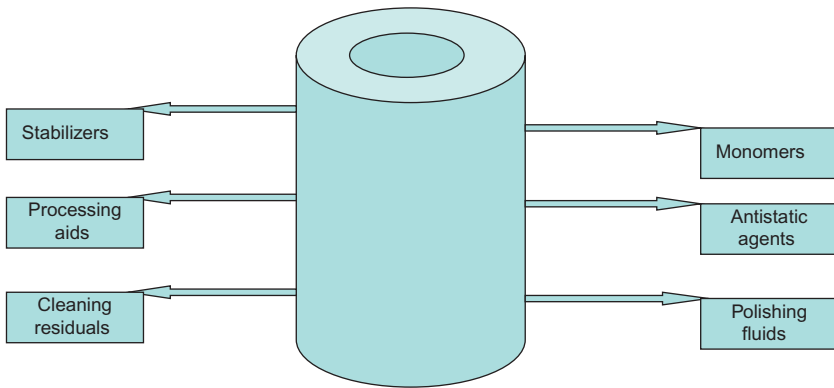


FIGURE 4.1 Examples of potential extractables and residues from polymeric biomaterials that could migrate into the surrounding environment.

Fouling of devices with even low levels of any of the above-mentioned contaminants can affect subsequent manufacturing processes as well, resulting in problems with adhesion, bonding or surface derivatization. The presence of surface films or particles, for example, interferes with the bonding of different parts in a medical device, and can lead to unacceptable levels of device failure [3]. Thus, cleanliness of medical devices is critical not only for patient safety but also for manufacturability. Conventional cleaning processes generally employ wet chemical techniques. Of these, the gentler approach may be the use of aqueous surfactant solutions to dissolve organic and inorganic contaminants [4]. While water is capable of dissolving many inorganic and some organic contaminants, not all residues dissolve readily in water. For this reason, most aqueous cleaners typically contain surfactants to physically reduce the surface tension of the solution. By reducing the surface tension, the surfactant aids in lifting the soil or contaminant away from the substrate being cleaned. Acids are also used to assist cleaning and to etch bonding surfaces. Unfortunately, despite the use of aggressive surfactants, highly purified deionized water, and several rinses, the solutions often leave residues in place of the original contamination. The new problem now is to remove the cleaner itself.

3 EXTRACTION TECHNIQUES

It is important to validate the extraction procedure for each device and contaminant(s). The extraction process must efficiently remove each target contaminant. For example, if the target residue is an oil, then water would most likely not be the solvent of choice, though it may be very useful when extracting acids. In some cases, devices or contaminants may be grouped into “families” which have similar characteristics [6,7]. Appropriate justification for these

groupings must be included in the validation file. Then the extraction validation is performed on each family [8].

The principle of “like dissolves like” comes into play during the extraction process. The cleaning solvent and the contaminant (soil) must spontaneously “like” each other. It is best if the solvent and the surface spontaneously like each other, too, because this allows the liquid to wet the surface lifting the soil into the bulk solvent [9]. Some potential contaminants from medical devices are water soluble, while others are soluble only in nonpolar solvents. For devices that may contact or have both water and non-water-soluble soils deposited on their surfaces, cleaning activity in both polar and nonpolar solvents is relevant. Conventional cleaning processes generally employ wet chemical techniques. Of these, the gentler approach is the use of aqueous surfactant solutions to dissolve organic and inorganic contaminants. Acids are also used to assist cleaning and to etch bonding surfaces. Unfortunately, despite the use of aggressive surfactants, highly purified deionized water, and several rinses, the solutions often leave residues in place of the original contamination.

The use of organic solvents overcomes some of the limitations of aqueous systems but introduces a different set of problems. Medical devices made from polymers that are sensitive to organic solvents can deform and lose the close tolerances needed for proper operation. While organic systems tend to leave fewer residues than aqueous systems, trace amounts of the solvent can cling and, consequently, lead to adverse device biocompatibility [10]. While requiring considerably less energy to evaporate than its aqueous counterpart, organic systems pose environmental hazards that must be addressed at a prohibitive cost. When wet chemical methods are ineffective in the removal of tenacious contaminants, sonication may be used to assist the cleaning process. The residual contamination, waste handling, and device damage problems associated with wet chemical techniques remain.

Extraction techniques are validated with spike recoveries, or exhaustive extractions, or both. Spike recovery uses positive controls with a known level of contamination. These controls are created by spiking clean devices with a known amount of target contaminant. Only a single extraction is needed to establish recovery efficiency. The spike recovery technique has a high degree of specificity because it demonstrates the removal efficiency of each contaminant or “family of contaminants” tested. While the spike recovery shows the removal efficiency of each contaminant tested, this method can get complicated as the number of contaminants on a device increase [8,10].

The exhaustive extraction technique uses positive controls which contain an unknown level of contamination. These “real-life” positive controls are created by exposing devices to the regular manufacturing operations but not to the cleaning process being validated. Positive controls for exhaustive extractions require repetitive extractions to establish recovery efficiency. The method provides “real-life” data but may not demonstrate adequate removal of every possible contaminant [8].

The exhaustive extraction method lets manufacturers establish precleaning levels of the target compounds, an important step in establishing limits for residual analysis. After establishing the precleaning levels, clean devices are tested to show a percent reduction of residual manufacturing materials. This percent reduction is an indication of the effectiveness of a cleaning process.

Medical components, such as hydrocephalus shunts, defibrillator leads, intraocular lenses, aortic grafts, and catheters contain reaction byproducts or residual raw materials that can be detrimental to the performance of the device or pose potential health risks in the body. For example, all silicone tubing and valves, parts which have been lightly cross-linked to provide tenacity, contain up to 5% leachable cyclic siloxanes; even “low-cyclic” content products that are cross-linked with platinum catalysts contain ~2% of the cyclic leachable species that can leach into the body over time [10].

Organic solvents, such as hexane and methylene chloride, could, in fact, be used to extract residual materials from silicone medical components, but remnants of these solvents can themselves be difficult to remove without degrading or altering the characteristics of the part. Leaving no solvent residues after extraction, supercritical CO₂ is extremely attractive for purifying silicone-based medical components. After the cyclics are extracted from a structure any absorbed CO₂ diffuses away when the pressure is reduced and the component retains its original shape and integrity [11].

4 CLEANING OF METAL AND CERAMIC COMPONENTS

Next-generation hip joints are being designed with porous surfaces that promote enhanced cell adhesion (Fig. 4.2), but the porosity presents a problem during cleaning operations.

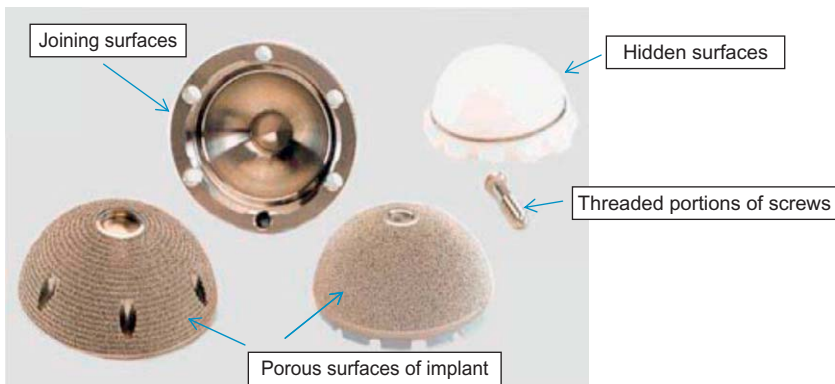


FIGURE 4.2 Examples of highly porous metallic implants that present a problem during cleaning operations.

Liquid solvents (isopropyl alcohol and aqueous surfactant solutions), because of their relatively high surface tension, cannot readily penetrate the pores to dissolve/extract the machining oils. Because supercritical CO₂ is, strictly speaking, a gas, it suffers no surface limitations, and thus, can penetrate into the smallest of pores and crevices and dissolve/extract the oils from orthopedic implants [11].

Supercritical CO₂ has proved to be successful in extracting machining oils; in fact, supercritical CO₂ cleans the titanium knee joint to a lower residual oil content than the standard multiple-flush process with aqueous surfactant solutions.

Residues from steam and hydrogen peroxide processes are composed of nontoxic chemicals, i.e., water and/or oxygen. For these processes, sterilant residues are not of concern. Other sterilization/disinfection processes, such as those using ethylene oxide (EO) and glutaraldehyde, may impart toxic residues. Additionally, reusable medical devices are sometimes made of polymeric or elastomeric materials, which may react with chemical sterilants resulting in the formation of chemical residues or extractables. EO and EO byproducts can be determined using the same procedures recommended for single use devices. Validation studies involving those processes for which residuals have not yet been characterized as toxic or nontoxic should include testing to determine whether toxic residues are present. If toxic residues are present, means to remove the residue (e.g., by aeration or rinsing) or to reduce it to nontoxic levels before patient use must be included in the sterilization/disinfection instructions. Establishing a nontoxic level may mean the use of toxicological risk assessment to establish an allowable limit for each chemical residue. This topic is discussed in [Section 7](#).

Methods outlined in Association for the Advancement of Medical Instrumentation (AAMI) ST29 recommend practices for determining residual EO in medical devices [12]. The International Standards Organization (ISO) has developed its EO Residual Standards and the FDA guidelines have been revised. ISO EO residue requirements will be different than current FDA or AAMI standards, in that ISO focuses on EO limits based on expected patient exposure potential versus the current AAMI/FDA approaches which measure actual EO concentration values taken at various time points poststerilization [2,12].

Since a reusable device will be subjected to multiple sterilant exposures, additional factors must be considered. Repeated processing may cause a buildup of EO or its byproducts ethylene chlorohydrin and ethylene glycol on the device. Increased aeration time may reduce residuals to low levels, or cleaning procedures employed between cycles may aid in the removal of residuals between resterilizations. It is important that residual analysis be conducted on devices that have experienced the full complement of reprocessing procedures expected during their useful life.

5 DEVICE COMPATIBILITY—TYPES OF TESTS AND DESCRIPTION

A factor to consider in choosing the extraction solvent is the compatibility with the device material(s). Simply stated, compatibility is the ability of two or more things to remain together with no after effects. There are some things we cannot change or control, such as the device or material to be cleaned or extracted, while the cleaning process, cleaning solvent and environment can be changed or controlled [13].

Cleaning and/or extraction solvents are likely culprits that can be the cause of material incompatibility. The amount and type of damage or change due to compatibility problems are diverse. Examples of change include swelling, softening, gain or loss of weight, embrittlement, crazing, dimensional changes and discoloration, to name a few.

Changes due to incompatibility may not always be obvious, or a change in a material may be a slow process such that over time there may be a deterioration of some critical property or function of the material or device [5].

Although the retention of functionality is recognized as one of the most pressing issues to deal with in a decision to sterilize and/or reuse, there is a scarcity of detailed scientific studies in the literature that have investigated the deterioration of materials and function in sterilized, reused, and resterilized medical devices.

Medical devices are commonly made of polymers or plastics of varying density. Low-density plastics are less resistant to heat than high-density ones. However, the properties of plastics are also achieved using additives which may stabilize the material to heat and light or reduce costs (fillers), enhance beneficial properties such as abrasion resistance, strength, lubrication, flexibility or antifungal properties. Some of these additives are susceptible to degradation as plastics used in the manufacture of medical devices are reprocessed, sterilized, and reused; while other additives can be altered through exposure to light [14]. Eventually, many of these degradation products can leach out of the device into the patient's body.

Mechanical testing can be used to help determine the functional integrity of devices. It may be hard to simulate in an experimental setting the various stresses a device may be exposed to in a clinical environment. Even if all the variables related to stress and strain were known, there is no one set of tests that are suitable for all devices. However, a key mechanical test is the tensile test which measures the force required to stretch a material through a range of extensions.

The most widely used instrument for stress-strain measurement is the Instron Tensile Tester. This instrument is essentially a device in which a sample is clamped between grips and jaws which are pulled apart at constant strain rates [14]. Tensile tests apply stress over a defined time scale. Tensile properties include the resistance of materials to pulling or stretching forces. The amount

of force required to break a material and the amount it extends before breaking are important properties. A variety of parameters are determined such as elongation, elongation at break, breaking strength, and tensile modulus of elasticity. The tensile modulus is defined as the ratio of stress to strain, and is determined from the initial slope of the stress-strain curve. The modulus of a material is a measure of the specimen's ability to resist deformation. Tensile modulus is also referred to as Young's modulus. Some plastic materials can be weakened by the process of sterilization or disinfection and therefore cause a decrease in the tensile strength of the material. Tensile strength data collected before and after cleaning, decontamination or sterilization can help predict any changes in the strength of the material [15,16].

Hardness testing is another useful mechanical test used to determine the functional characteristics of material used in a medical device or instrument. Hardness is defined by ISO as an indication of the resistance of a material to indentation, scratch resistance, and/or rebound resilience. ISO standards report three methods for measuring hardness: a Shore hardness, a ball indentation hardness, and Rockwell hardness [14,17]. It is important to emphasize that hardness values obtained from one method in general cannot be compared with those derived from another, although data can be empirically compared. Generally, hardness is used as an end-performance property of material used in a device. The instrument used to obtain the measurement is called a durometer.

Thermal analysis, the response of a polymer to controlled heating processes, is a family of techniques widely used in the development and characterization of materials, including plastics and elastomers. Characterizing a polymer's glass transition temperature, melting point, and extent of crystallinity are important and often used to produce materials whose properties are tailored to the product's ultimate application [18]. The primary thermal analysis techniques for certifying product quality are differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA). Thermal analysis tests for certifying product quality include those shown in Table 4.1.

DSC results can give rapid measures of thermal characteristics, such as the polymer melting point and glass transition temperature. The nature of these

TABLE 4.1 Thermal Analysis Techniques for Materials

Thermal Property	Technique
Melting point	DSC
Degree of crystallinity	DSC
Glass transition temperature	DSC, TGA
Component quantification	TGA

transitions, in addition to identifying characteristics unique to each polymer, can also provide information about its phase structure, thermal history, and purity. The loss of polymer additives can have a dramatic effect on these thermal properties.

6 TEST METHODS TO IDENTIFY CONTAMINANTS IN EXTRACTS

6.1 Chromatographic Analysis

The various chromatographic methods, such as gas, liquid, paper, and thin-layer, have become indispensable aids in the isolation, separation, and detection of chemicals. Gas and liquid chromatography, especially high-performance liquid chromatography (HPLC), have become powerful analytical tools to characterize additives in polymeric materials used in medical devices as well as disinfectants and liquid sterilants adsorbed onto surfaces [15]. Gas chromatography and gas chromatography-mass spectrometry (GC/MS) are commonly used for detection and quantitation of detergent residue. These methods provide separation, identification and quantitation of results when an acceptable reference standard is used.

HPLC is undeniably one of the fastest growing and useful among all the analytical separation techniques. The reason for this growth is attributable to the sensitivity of the method, its ready adaptability to accurate quantitative determinations, and its suitability for separating nonvolatile or thermally unstable species. Liquid chromatography, because of its great flexibility and widespread applicability, can be used for the analysis of over 80% of all known organic compounds [14].

6.2 Spectrophotometric Analysis

6.2.1 Infrared

Infrared (IR) spectroscopy measures the vibrational spectrum of a sample by passing IR radiation through it and recording which wavelengths have been absorbed and to what extent. Since the amount of energy absorbed is a function of the number of molecules present, the IR spectroscopy provides both qualitative and quantitative information. Since the IR spectrum of a chemical compound is perhaps its most characteristic physical property, IR spectroscopy finds extensive application in identifying substances and their respective concentrations. Many cleaning and disinfecting agents have distinctive IR spectra that can be used to identify residuals on the surface of devices and instruments [19,20].

If residual glutaraldehyde is present on a medical device, then it can be extracted in a solvent and the solvent analyzed for glutaraldehyde. Many detergents and enzymes (proteins) can be identified by this analytical method.

6.2.2 *Ultraviolet/Visible Spectroscopy*

Much like IR, ultraviolet (UV) and visible spectra of a chemical can be used to identify and determine the concentration of analytes in an extract. Enzymes, which are proteins, are very easily detected and their concentration determined by their distinct UV spectra [15]. Most proteins absorb maximally at 280 nm. The method provides quantitative results, offers fast spectral acquisition and is not limited to water as the extraction solvent. But it does lack peak separation and requires a chromophore for specificity.

6.3 Total Organic Carbon

Total organic carbon (TOC) analysis is specific to organic compounds and theoretically measures all the covalently bonded carbon in water. This method has been recognized and accepted as a valid way to detect residues of contaminants. It is a very sensitive method with detection capabilities in the parts-per-billion range [8].

TOC analysis does have some limitations and drawbacks. Contaminants must be water soluble and it must be established that a substantial amount of the contaminating material(s) is organic and contains carbon that can be oxidized under TOC conditions. Excellent water quality, such as USP “Purified Water” is needed for sensitivity and to reduce interfering substances. By coupling TOC with a conductivity/pH analysis, device extracts may also be analyzed for any ionized compounds such as acids, bases, or salts.

6.4 Gravimetric Analysis

A basic gravimetric analysis for medical devices is described in ASTM F2459 [21]. This method quantifies any extractable residue that can be removed by the chosen solvent. One advantage of this method is that extraction solvents other than purified water can be used. However, the method excludes residues that are more volatile than the extraction solvent. The sensitivity of the method is dependent on the capability of the analytical balance used. The detection capabilities are usually between 0.01 and 0.1 mg.

The residue obtained from the gravimetric analysis can also be characterized by IR spectroscopy. The spectrum of the residue can reveal the presence of certain compounds such as hydrocarbons, silicones, amines, etc. Identification can also be accomplished by comparing the residue spectrum to the spectra of the target compounds.

7 SETTING LIMITS AND BIOLOGICAL SAFETY ASSESSMENT

One of the greatest challenges in medical device chemical characterization is performing adequate assessment of biological or toxicological risks from residues, including cleaning and sterilization residues that can compromise patient

safety. There are no established regulatory limits for residual analysis, so it is important to justify an acceptable level. The primary reason for this is due to the number of variables involved. Every contaminant has a differing degree of toxicity, and this toxicity can change depending on where and how long it is exposed to the patient. Therefore, regulatory agencies have left it up to individual companies to establish and justify residue limits [22].

A few practical techniques for setting limits include comparing clean and unclean devices, evaluating a new process versus an old one, and assessing residue amounts based on the detection limit of the method. Residue limits are usually specified in micrograms or milligrams per device. When comparing devices with different surface areas, it is advisable to define limits based on surface area, such as $\mu\text{g}/\text{cm}^2$ or mg/cm^2 . By expressing the residue limit based on surface area, the same limit may be used for devices of differing sizes.

Probably, the most common technique used to justify residue limits is performing biocompatibility testing. Demonstrating the biocompatibility of the clean devices means that any residues on that clean device are also biocompatible. *In vitro* cytotoxicity is a very sensitive method because the device extracts are applied directly to the cells. The test is scored based on the degree of cellular destruction [3]. This test has defined pass/fail criterion found in USP <87>.

Another approach that has gained popularity is the use of toxicological risk assessments to set the allowable limit of a residual based on its toxicological profile. ISO 10993-17 has provided a working standard that can be used to establish the allowable limits of leachables, extractables, and manufacturing residues as an essential part of medical device biocompatibility [23]. When properly implemented as part of the overall cleaning process, the biological risk assessment becomes a powerful tool in the establishment and assurance of biological safety [8]. Toxicological hazard is a property of the chemical residues left on a medical device as a result of manufacturing and should be considered in relation to the assurance of biological safety. Therefore, for a biological safety assessment, the first step comprises characterization of residual contaminants. Residues may be analyzed by direct surface analysis or measured after extraction. Extraction analysis is most common so the methods discussed here are all extraction based. Toxicological hazards can be identified from knowledge of the toxicity of materials or extracted chemicals.

A systematic analysis of biological risks is required using the general principles set out in clause 3 of ISO 10993-1 [24]. Unfortunately the matrix in ISO 10993-1 is often used as a check list to perform a standard set of tests. What is actually needed is an appropriate scientific evaluation program based on the specifics of the device.

The results of all tests should be interpreted in the context of the overall risk assessment to know whether a specific outcome indicates acceptable risk or not. This collaborative approach emphasizes the need for an overall scientifically valid risk assessment.

The suitability of a medical device for a particular use involves balancing any identified risks with the clinical benefit to the patient associated with its use. ISO 10993-17 states that “among the risks to be considered are those arising from exposure to leachable substances arising from medical devices.” This standard provides a method for calculating maximum tolerable levels that may be used by “other standards-developing organizations, government agencies, and regulatory bodies. Manufacturers and processors may use the allowable limits derived to optimize processes and aid in the choice of materials in order to protect patient health.”

The aim of the assessment should be to identify any biological hazards inherent in the materials used in the medical device and to estimate the risks resulting from these in light of the intended use. The goal is to develop a process that ultimately protects public health and establishes the safety of medical devices. This objective is supported by ISO 10993-17 in subclause 4.3 of the general principles for establishing allowable limits which states that “the safety of medical devices requires an absence of unacceptable health risk.” The manufacturer of a medical device is responsible for assuring its biological safety and for documenting the assessment of toxicological risks and establishing the effectiveness of the analysis. Evidence must be provided that an appropriate toxicological risk assessment has been carried out so that it can ensure that public health is not endangered. ISO 10993-17 also adds that “where risk associated with exposure to particular leachable substances are unacceptable; this part of ISO 10993 can be used to qualify alternative materials or processes.” This is another example of the way risk assessment can be used as a tool for critical decision making.

Although significant risks arising from hazardous residues are identified by chemical characterization, their acceptance should be assessed in line with established toxicological principles. Biocompatibility tests identified in the ISO 10993 series of standards may be used to provide further assessment of risk.

The value of risk assessments has long been recognized by international organizations and now by ISO 10993. In the risk assessment process, a decision must be made related to risk versus benefit. Once risk assessment has been completed, the focus turns to risk management. Decisions should be made based on the results of risk assessment, biological safety testing and safe clinical use of predicate devices as described in ISO 14971 [25]. When coupled or linked to biological safety testing, a successful biological safety assessment becomes a highly useful decision-making tool.

8 VALIDATION

In addition to minimizing contamination on medical devices, it is crucial to develop, define, validate, and control the cleaning process. There are many sources of contamination. Inevitably, assembly activities, process chemistries, and process materials can contribute to contamination; therefore, it is important

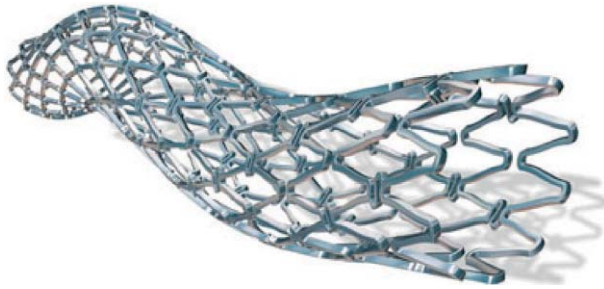


FIGURE 4.3 Bare metal stents used in coronary artery disease. This is an example of a medical device that can cause challenges in evaluating the cleaning process.

to physically remove undesirable surface material regardless of its origin. It has been said that contamination is nothing more than matter out of place and that cleaning is the removal of that matter. Cleaning is the removal of residues, contaminants, or soils—unwanted or extraneous material, deposited on or attached to a substrate surface. Devices with complex geometries are more challenging with respect to evaluating the cleaning process (Fig. 4.3). No matter how you define it or examine it, it is unwanted matter. With sterilizing or disinfecting, the issue is rendering the contaminant nonviable. With cleaning, the issue is physically removing contaminants from the surface without altering that surface in a way that interferes with product performance. For real-world uses in the medical device industry, cleaning includes removing soil and achieving the appropriate level of cleanliness, surface quality, and surface characteristics.

9 SOURCES OF CONTAMINATION

Sources of viable contaminants can vary. The number of sources can be best illustrated in a figure, which also provides a useful summary for sources of nonviable contamination, as shown in Fig. 4.4.

There is no greater challenge to device manufacturers than that of controlling viable and nonviable contaminants that originate from the human body. Devices that are assembled usually by hand without protective garments present a critical environmental challenge. Using protective clothing while conducting routine production activities can result in decreased particulates (viable and nonviable) generated from the human body. Wherever the process can be automated, contamination from the human element can be eliminated. However, successful automation requires a high degree of understanding of process variables; in low production situations, people may continue to assemble products. In general, training, education, and engagement of the workforce as to the importance of the product and manufacturing process promote contamination control. A responsible technician can also contribute to quality control.

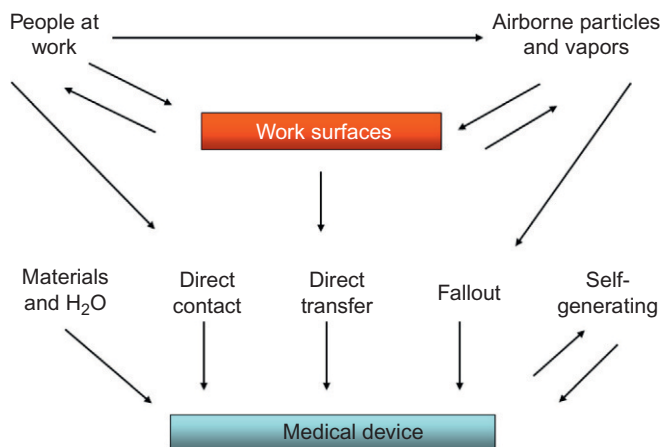


FIGURE 4.4 Sources of viable contamination.

10 CLEANLINESS STANDARDS, NONVIALE CONTAMINATION

Defining the acceptable level of nonviable contamination is very device dependent. This is, in part, because of the diversity of contaminants and, in part, due to our incomplete understanding of desirable surface characteristics and attributes. The development of general cleanliness and contamination standards has been gradual; however, there are a few standards that can be useful with regard to the evaluation and interpretation of nonviable contamination. Some of them are discussed further.

10.1 Setting Allowable Limits

Part of the series covering “Biological Evaluation of Medical Devices” ISO 10993-17 is a normative standard covering “Methods for the Establishment of Allowable Limits for Leachable Substances” [23]. It is a normative (prescriptive) standard of utility to those involved in both initial manufacture and in the reprocessing of medical devices. ISO 10993-17 can be used as part of cleaning validation in terms of hazard assessment, to identify potential contaminants or residues and their associated risks. The standard also covers exposure assessment, including selection of test and extraction methods, establishment of acceptance criteria, and method validation. The standard is particularly useful in characterizing the risk based on hazard identification, hazard assessment, and exposure assessment. It is critical for the company to document the rationale used in the assessment and to scrupulously maintain pertinent records.

10.2 Extraction and Gravimetric Analysis

ASTM F2459-05, “Standard Test Method for Extracting Residue for Metallic Medical Components and Quantifying via Gravimetric Analysis,” is a method for determination of weight of residue recovered from extract(s) of medical device components [21]. One might expect additional, more specific methods to gradually appear.

10.3 Quality Management Systems

ISO 13485:2003 and Federal Regulation 21CFR820 are complementary global and United States standards [26,27]. As indicated previously, when fabrication of complex devices occurs at multiple locations, adoption of quality management systems (QMS) compels a renewed emphasis on cleaning and contamination control. QMS also leads to increased transparency among companies. Risk management programs are not restricted to the final assembler of the devices. This means that process chemistries, cleaning chemistries, and cleaning processes will need to be better understood at all levels of production.

11 FUTURE DEVELOPMENTS

A methodical approach for understanding and applying standards, cleaning processes testing, and establishing a routine monitoring program for contaminants are key elements in validating an environmentally controlled manufacturing facility. In the last few years, challenges to the medical device manufacturing community have increased and so have performance expectations. And with the ever-decreasing size of medical devices, these expectations are becoming more difficult to meet.

Residues on the surface of medical devices can cause implant failure and poor device performance. The main source of these residues is from materials used in the manufacture of the device, although contamination during the storage, cleaning and handling of the device is also known to occur. Small amounts of these surface residues can cause deleterious effects in patients, because the residues are in direct contact with body tissues.

In order to minimize contamination, the FDA stipulates that medical device manufacturers follow specific cleanliness validation procedures. First, they must identify all possible residues present on the device and set an acceptable residue limit. Then, they must use a cleaning regime that reduces residue levels below this limit, without leaving significant levels of cleaning agent behind. Finally, documentation to verify that residue limits are not exceeded must be submitted to the FDA before the device can go on the market.

Part of the problem for medical device manufacturers and device cleaners is that there are no official residue limits or specified analytical techniques to measure residue levels. Therefore, the manufacturers have to base their judgments

on existing medical devices on the market, which may not be possible if the device has a novel use or uses different materials or manufacturing processes to devices currently on the market. Another issue is that the surfaces of medical devices are becoming more complex in terms of their geometry and chemistry, making cleaning more difficult.

Because processes are so complex and have so many variables, many standards are normative (prescriptive) by necessity. Therefore, it becomes important for the device manufacturer to develop a system involving external standards, company-specific standards and controls, and on-going monitoring/vigilance. The entire supply chain, including suppliers of process chemicals, must be involved in a coordinated effort. Controlling nonviable and viable contaminants includes determining the appropriate product specifications and the level of cleanliness to be maintained. Such control means establishing alert and action limits, conducting routine monitoring, and obtaining (and when warranted, acting upon) trending data.

To help make the best choices in controlling and minimizing risks to critical product from viable and nonviable contaminants, it is important to have frank and open communications with outside advisors, including consultants and analytical testing facilities. Choosing the optimal testing protocols, interpreting results, selecting the appropriate chemistries and implementing rugged, reliable processes requires experienced advice and input.

Organizations like the American Society for Testing and Materials (ASTM) are very active in the medical device industry. In fact, ASTM Committee F04 on Medical and Surgical Materials and Devices is a standards-setting group that meets regularly and holds worldwide seminars and meetings with the explicit mission of setting standards for the medical device industry. This organization has representation from industry as well as regulatory agencies like the FDA and Notified Bodies of Europe. While quite active today, ASTM F04 is expected to be even more involved in the development of standards that set limits for cleaning residues in the future.

Another expected change with regard to cleaning is the development of new and more sensitive analytical equipment and methods that can detect smaller amounts of residue on smaller devices. As devices become smaller, the challenge increases for methods and instruments that can detect and analyze trace amounts of contaminant.

12 CONCLUSION

Cleaning, during the manufacturing process, prepares medical devices for sterilization and sterile packaging. Successful cleaning, decontamination and sterilization/disinfection of medical devices and instruments require both the careful selection of materials and consistent monitoring of the procedures used to process them. Usually medical devices are made of fairly robust materials that can withstand corrosive blood and body fluids. However, in some instances,

one portion of the device is intended to come into contact with skin, blood, or body fluids, while other parts may remain exterior to the human body and may be less sturdy.

Chemical characterization of residues that may be adsorbed onto surfaces and mechanical testing to ensure functionality should provide sufficient information to evaluate the potential success of cleaned and disinfected/sterilized medical devices. By using a combination of chemical and mechanical analysis techniques, both manufacturing and decontamination processes can be optimized to ensure a safe and effective product. It is important to note that no test, however foolproof its design, can ever be considered a definitive predictor of clinical performance. Whichever scientifically based, predictive testing procedure is used, it is very difficult to devise a test system that accurately replicates the conditions with the human body, so predictiveness is never going to be guaranteed.

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